

for the above example:

```
((- 30 100) 10 20) ; 30 - 100 = -70
((- 20 (-70)) 10) ; 20 - (-70) = 90
(- 10 90) ; 10 - 90 = -80
```

The ‘-reduce’ function applies the function to the first two elements of the list, then applies the function to the previous result, and then to the third element of the list, and so on. For example:

```
(-reduce function list) ;; Syntax

(-reduce '- '(1 2 3 4))
-8
```

The function application evaluation order for the above example is as follows:

```
((- 1 2) 3 4) ; 1 - 2 = -1
((- (-1) 3) 4) ; (-1) - 3 = -4
(- (-4) 4) ; (-4) - 4 = -8
```

The ‘-reduce-r’ function is similar to the ‘-reduce’ function with the exception that it evaluates from right-to-left. An example to illustrate the function is given below:

```
(-reduce-r function list) ;; Syntax

(-reduce-r '- '(1 2 3 4))
-2
```

The function application evaluation order is as follows for the above example:

```
((- 3 4) 1 2) ; 3 - 4 = -1
((- 2 (-1)) 1) ; 2 - (-1) = 3
(- 1 3) ; 1 - 3 = -2
```

The ‘-count’ function takes a predicate function and a list as arguments. It counts the number of items that satisfy the predicate function. For example:

```
(-count predicate list) ;; Syntax

(-count 'even? '(1 2 3 4))
2
```

The ‘-sum’ function simply returns the sum of the list. A few examples are shown below:

```
(-sum list) ;; Syntax
```

```
(-sum '())
0
(-sum '(1))
1
(-sum '(1 2 3 4))
10
```

The ‘-product’ function computes the product of the elements in a list. An example is given below:

```
(-product list) ;; Syntax

(-product '(1 2 3 4))
24
```

The ‘-min’ and ‘-max’ functions return the smallest and largest values from the items in a list. A couple of examples are shown below:

```
(-min list) ;; Syntax
(-max list) ;; Syntax

(-min '(5 2 3 4))
2

(-max '(1 2 3))
3
```

The ‘-min-by’ and ‘-max-by’ functions take a comparator function as an argument, and return the least and greatest elements in a list, based on the comparison function. A few examples are given below:

```
(-min-by comparator list) ;; Syntax
(-max-by comparator list) ;; Syntax

(-min-by '>' '(5 2 3 4))
2

(-max-by '>' '(5 2 3 4))
5

(--max-by (> (length it) (length other)) '((1 2 3) (4) (5 6)))
(1 2 3)
```

We will explore more functions from dash.el in the second part of this series of articles. **END** 

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